

A Sharp Negative Answer and Classification for Schwartz-Window Gabor Frames over Rational Lattices

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Status. `literature_already_answered`. This note identifies an exact later resolution of the question in arXiv:2408.03423; it does not claim a new theorem. The answer is negative in general, and the later paper gives a sharp if-and-only-if replacement.

The source question and its partial answer

For a full-rank phase-space lattice $\Gamma \subset \mathbb{R}^{2d}$, Enstad, Thiel and Vilalta ask whether

$$\text{covol}(\Gamma) < 1 \tag{1}$$

always implies the existence of $g \in \mathcal{S}(\mathbb{R}^d)$ such that the Gabor system $\{\pi(\gamma)g : \gamma \in \Gamma\}$ is a frame [1, Problem A]. The unresolved part is $d \geq 2$ with Γ rational for the symplectic form.

Write

$$\Gamma_{\text{int}} = \{\gamma \in \Gamma : \Omega(\gamma, \gamma') \in \mathbb{Z} \text{ for every } \gamma' \in \Gamma\}, \quad n_\Gamma = [\Gamma : \Gamma_{\text{int}}]^{1/2}.$$

The source proves existence under the stronger condition

$$\text{covol}(\Gamma) < 1 - \frac{d-1}{n_\Gamma}. \tag{2}$$

It explicitly reports that no Schwartz frame was then known for a lattice violating (2).

Exact later classification

For the symplectic adjoint lattice Γ° , put $\nu(\Gamma) = [\Gamma : \Gamma \cap \Gamma^\circ]^{1/2}$ whenever this index is finite. Thus, on the rational lattices of the source, $\nu(\Gamma) = n_\Gamma$. Caragea, Kolountzakis and Pfander prove the following in a preprint posted August 7, 2026 [2, Theorems 1.2 and 1.3].

Theorem (Caragea–Kolountzakis–Pfander). *A full-rank lattice $\Gamma \subset \mathbb{R}^{2d}$ admits a Schwartz function g generating a Parseval Gabor frame if and only if*

$$\text{covol}(\Gamma) < 1 \quad \text{and, when } \Gamma \text{ is symplectically rational, } \nu(\Gamma) \geq \nu(\Gamma^\circ) + d. \tag{3}$$

If either inequality fails, no window in the Feichtinger algebra $S_0(\mathbb{R}^d)$ can generate a Gabor frame.

This is stronger on the nonexistence side because $\mathcal{S}(\mathbb{R}^d) \subset S_0(\mathbb{R}^d)$. It also gives Parseval windows on the existence side, which of course are frames.

Why the classification exactly sharpens the source theorem

The later paper proves, for a rational lattice,

$$\text{covol}(\Gamma) = \frac{\nu(\Gamma^\circ)}{\nu(\Gamma)}. \quad (4)$$

Set $n = n_\Gamma = \nu(\Gamma)$ and $m = \nu(\Gamma^\circ)$. Since m, n are integers, the source inequality (2) is equivalent to

$$\frac{m}{n} < 1 - \frac{d-1}{n} \iff m < n - d + 1 \iff m \leq n - d \iff n \geq m + d.$$

This is exactly (3). Thus the source's sufficient condition was already the sharp one; the later theorem supplies the missing necessity.

Concrete counterexample to the broad question

Take $d = 2$, use coordinates $(x_1, x_2, \omega_1, \omega_2)$, and let

$$\Gamma = \text{diag}(1, 1, 1, \tfrac{1}{2})\mathbb{Z}^4.$$

Then $\text{covol}(\Gamma) = 1/2 < 1$, while the later paper computes

$$\nu(\Gamma) = 2, \quad \nu(\Gamma^\circ) = 1.$$

Condition (3) would require $2 \geq 1 + 2$, which is false. Therefore no $g \in S_0(\mathbb{R}^2)$, and in particular no Schwartz function, generates a Gabor frame over this lattice. This single lattice gives a full negative answer to Problem A. More generally, (3) classifies every rational and non-rational lattice.

Verification scope

The included PDFs and TeX sources record both sides of the implication. In the source, Problem A, Definition B, Theorem C, and the concluding remarks give the question and (2). In the later preprint, Theorems 1.2, 1.3 and 1.4, Examples 1.5(ii), and equation (1.8) give the classification, the stronger S_0 obstruction, the displayed counterexample, and identity (4). Because [2] is a very recent arXiv preprint, this packet records an exact literature resolution rather than an independently refereed verification of its long proof.

References

- [1] U. Enstad, H. Thiel and E. Vilalta, *Criteria for the existence of Schwartz Gabor frames over rational lattices*, Int. Math. Res. Not. IMRN (2025), rnaf038; [arXiv:2408.03423](#).
- [2] A. Caragea, M. N. Kolountzakis and G. E. Pfander, *Tilings, packings, and the existence of Schwartz-class Gabor windows*, [arXiv:2608.06679](#), August 2026.